

Basic Concepts

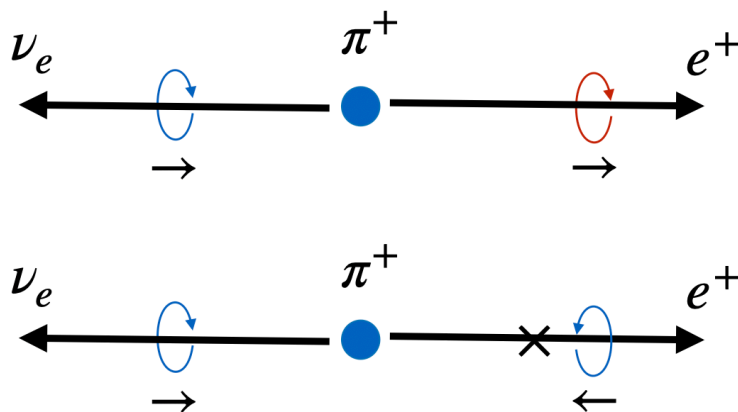
Standard Model and Fermion Masses

These lectures are part of the Beyond the Standard Model course I taught at the University of Iowa in Fall 2025. This part of the course was designed to be a conceptual introduction to the physics of neutrino masses. You will benefit most from these notes if you have already taken at least some Quantum Field Theory. They are very much not self-contained and a work in progress.



Some useful references for further study include:

- S.M. Bilenky and S. Petcov, Rev. Mod. Phys. 59, 671 - Published 1 July, 1987, <https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.59.671>
- Introduction to the Physics of Massive and Mixed Neutrinos, S.M. Bilenky, Lecture Notes in Physics, 2010, <https://link.springer.com/book/10.1007/978-3-319-74802-3>
- Carlo Giunti, Chung W. Kim, Fundamentals of Neutrino Physics and Astrophysics, Oxford University Press, 2007, http://theor.jinr.ru/~vnaumov/Eng/JINR_Lectures/books/Giunti2007.pdf
- H. Georgi, Weak Interactions and Modern Particle Theory, Benjamin/Cummings, 1984, <https://www.physics.mcgill.ca/~jcline/742/georgi-weak-interactions.pdf>.



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1 Basics and Dimensional Analysis

We start with some basics facts about the Standard Model (SM) and some dimensional analysis. Throughout these lectures, we will use natural units where $\hbar = c = 1$ and the metric signature is $(+, -, -, -)$.

The Standard Model is a gauge theory with a Lagrangian formulation. The action S is a functional of the fields and is defined as the integral of the Lagrangian density $\mathcal{L}$ over spacetime:

$$S = \int dt L = \int d^4x \mathcal{L}. \quad (1.1)$$

In natural units, the action is dimensionless (it appears as a phase $e^{iS/\hbar}$ in the path integral). By Eq. (1.1), the Lagrangian density must have mass dimension 4:

$$[S] = E^0, \quad [d^4x] = E^{-4}, \quad [\mathcal{L}] = E^4, \quad (1.2)$$

where $[O]$ denotes the mass dimension of the operator/field/parameter O in natural units. To be clear, the mass dimension of a mass, like $m_\mu \sim 105$ MeV, is E and of a dimensionless coupling, like $\alpha = 1/137$, is E^0 .

The mass dimension of different types of fields is determined by the requirement that the kinetic term in the Lagrangian has mass dimension 4. For example, for a scalar field ϕ , the kinetic term is

$$\mathcal{L}_{\text{kinetic}} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi), \quad (1.3)$$

and so we must have $[\phi] = E^1$. Recall that $[x^\mu] = E^{-1}$, so $[\partial_\mu] = E^1$. Similarly, for a fermionic field ψ , the kinetic term is

$$\mathcal{L}_{\text{kinetic}} = \bar{\psi} i \not{\partial} \psi = \bar{\psi} i \gamma^\mu \partial_\mu \psi, \quad (1.4)$$

and we find that $[\psi] = E^{3/2}$. Note that ψ is a four-component spinor, and the γ^μ matrices are dimensionless.¹ Finally, for a gauge field A_μ the kinetic term is

$$\mathcal{L}_{\text{kinetic}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (1.5)$$

we find that $[A_\mu] = E^1$. In fact, all fermions have mass dimension $3/2$ and all bosons have mass dimension 1 in four spacetime dimensions. You can prove this from the commutation relations. For example, for a scalar field, $[\phi(\mathbf{x}), \pi(\mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y})$ with ϕ the scalar field and $\pi = \dot{\phi}$ its conjugate momentum, and $[\delta^3(\mathbf{x})] = E^3$. For fermions, one uses $\{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = \delta_{\alpha\beta}\delta^3(\mathbf{x} - \mathbf{y})$.

The Lagrangian is a complex number, so all terms in $\mathcal{L}$ must also be numbers and must, individually, have mass dimension 4. If a given term in our Lagrangian has mass

¹In what follows, I will always omit the spinor indices, but note that the kinetic term is a contraction of both spinor (α) and Lorentz (μ) indices: $\bar{\psi} i \not{\partial} \psi = \bar{\psi}_\alpha i (\gamma^\mu)^{\alpha\beta} \partial_\mu \psi_\beta$. An individual Dirac *matrix*, say $\gamma_{\alpha\beta}^0$, is a matrix in spinor space, but $\gamma_{\alpha\beta}^\mu$ is a Lorentz vector of (spinor) matrices. In the end, the kinetic term is just a complex number, i.e., a Lorentz scalar.

dimension other than 4, then we must multiply it by some mass scale Λ to make the mass dimensions work out:

$$\mathcal{L} = \frac{C}{\Lambda^{d-4}} \mathcal{O}_d, \quad [\mathcal{O}_d] = E^d, \quad [C] = E^0, \quad [\Lambda] = E^1, \quad (1.6)$$

So, C is just a number and Λ is some unknown, but presumably measurable mass scale. For example, if $d = 5$, the operator $\mathcal{O}_5$ is suppressed by a single power of the mass scale Λ . This is called a dimension-5 operator, and we often assume $C = 1$ for simplicity:

$$\mathcal{L} = \frac{1}{\Lambda} \mathcal{O}_5. \quad (1.7)$$

Like any other term put by hand in the Lagrangian, the coefficient of a dimension-5 operator is a free parameter that must be determined by experiment. The difference here is that this coefficient is not dimensionless, like $\alpha = 1/137$ or α_s . Measuring the coefficient of such a dimension-5 operator in our Lagrangian amounts to inferring the mass scale Λ (more precisely, Λ/C). Typically, this scale is much larger than the energies we can probe directly, and so the larger the dimension of the operator, the more suppressed its effects are at low energies.

This idea leads to an expansion in inverse powers of Λ :

$$\mathcal{L} = \mathcal{L}_{\text{dim-4}} + \sum_i \frac{C_i}{\Lambda} \mathcal{O}_d^i + \sum_j \frac{C_j}{\Lambda^2} \mathcal{O}_d^j + \dots, \quad (1.8)$$

This is known as an effective field theory (EFT) expansion. The key assumption is that the higher the dimension of the operator, the smaller the physical effect of that operator at low energies is. So dimension-8 operators should be less relevant than dimension-6 operators, which should be less relevant than dimension-5 operators, and so on.

We can make connection with a very familiar example: the Fermi theory of weak interactions. At low energies, beta decay can be accurately described by a four-fermion interaction

$$\mathcal{L} = -\frac{G_F}{\sqrt{2}} \bar{p} \gamma^\mu (1 - \gamma^5) n \bar{e} \gamma_\mu (1 - \gamma^5) \nu_e = -\frac{1}{\Lambda^2} O_6, \quad (1.9)$$

where $G_F/\sqrt{2} = g^2/(8m_W^2)$ is the Fermi constant. In this example, the fermion operator has dimension six, $O_6 = \bar{p} \gamma^\mu (1 - \gamma^5) n \bar{e} \gamma_\mu (1 - \gamma^5) \nu_e$, and we can identify $\Lambda \sim m_W$ up to coupling factors and so on. This higher-dimensional operator arises as the small momentum transfer limit of the W -boson propagator in the full electroweak theory. This is also the limit in which the propagator for the W boson shrinks down to a point, giving this approximation the name of *contact interaction* (this language is equivalent to saying the entire beta decay process is mediated by a local point-like interaction). Diagrammatically:

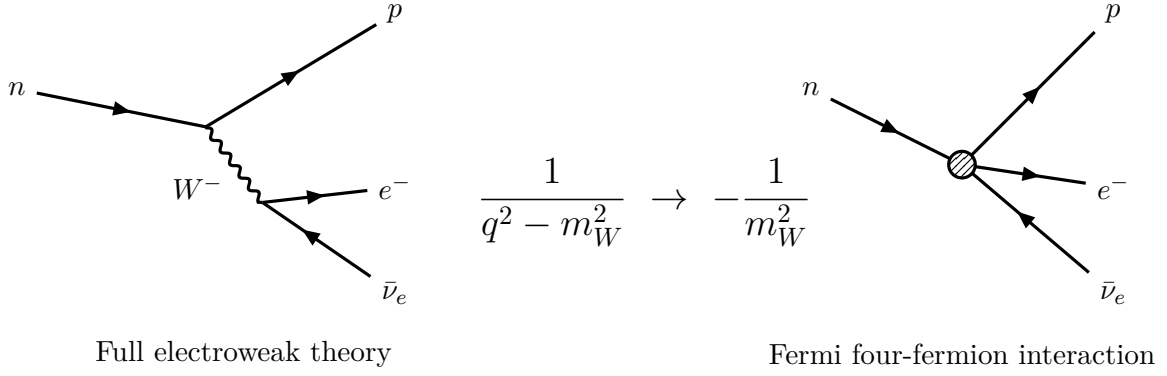


Figure 1: For small momentum transfers $q^2 \ll m_W^2$, the virtual W^- propagator in neutron beta decay can be replaced by a local four-fermion operator.



In non-relativistic quantum mechanics, one can see the same limit by looking at the Yukawa potential generated by the exchange of a massive scalar particle ϕ of mass m_ϕ . As m_ϕ gets big:

$$V(r) = -\frac{g^2}{4\pi} \frac{e^{-m_\phi r}}{r} \xrightarrow{m_\phi r \gg 1} -\frac{g^2}{m_\phi^2} \delta^3(\vec{r}), \quad (1.10)$$

where g is the coupling of the scalar to the fermions. In this limit, the range of the interaction is very short, and the potential is well approximated by a contact/local interaction, just like weak interactions at low energies. Note how, in addition to being short-ranged, the contact potential is suppressed by the mass of the mediator, m_ϕ .



Theories with higher-dimensional operators ($d > 4$) are non-renormalizable. The SM is a renormalizable theory, but we will see that its simplest extension that accommodates neutrino masses is a dimension-5 operator. In this case, we treat the SM as an effective field theory (EFT), also known as SMEFT, valid up to some energy scale Λ where new physics appears. The lack of renormalizability means that the theory is only valid up to a certain energy scale, beyond which new physics must appear; it also means that any given operator requires its own counterterm to cancel divergences, so the theory has an infinite number of free parameters. Every operator requires its own measurement to fix its coefficient. You can learn more about EFTs in the following references:

- A. Manohar, *Introduction to Effective Field Theories*, <https://arxiv.org/abs/1804.05863>
- D. B. Kaplan, *Effective Field Theories*, https://archive.int.washington.edu/users/dbkaplan/EFT_lectures.pdf
- T. Cohen, *As Scales Become Separated: Lectures on Effective Field Theory*, <https://arxiv.org/abs/1903.03622>

The Standard Model Gauge Group: The SM Lagrangian is invariant under the following gauge group:

$$G_{\text{SM}} = SU(3)_C \times SU(2)_L \times U(1)_Y, \quad (1.11)$$

where $SU(3)_C$ is the gauge group of the strong interactions, $SU(2)_L$ is the gauge group of the weak interactions, and $U(1)_Y$ is the gauge group of the hypercharge interactions. The fermion fields are said to be *charged* under these gauge groups. Mathematically, the charge is defined as the representation of the field under the gauge group. If the field transforms trivially under the gauge group, it is said to be a singlet and chargeless under that group (for example, right-handed particles are singlets/chargeless under $SU(2)_L$ and leptons are singlets/chargeless under $SU(3)_C$).

Each symmetry group factor comes with its own gauge bosons. The number of generators of each group determines the number of gauge bosons. There are $N^2 - 1$ generators for $SU(N)$ and one generator for $U(1)$, so in the SM:

$$SU(3)_C : 8 \text{ gluons } g_\mu^a, \quad a = 1, \dots, 8 \quad (1.12)$$

$$SU(2)_L : 3 \text{ weak bosons } W_\mu^i, \quad i = 1, 2, 3 \quad (1.13)$$

$$U(1)_Y : 1 \text{ hypercharge boson } B_\mu. \quad (1.14)$$

From these, we define the physical $W^\pm$ bosons as $W_\mu^\pm = (W_\mu^1 \mp iW_\mu^2)/\sqrt{2}$. The neutral electroweak gauge bosons W_μ^3 and B_μ will eventually mix to form the physical Z_μ boson and the photon A_μ .

The simplest and most unique field in the SM is the Higgs field. It is responsible for giving mass to the fermions and gauge bosons through the mechanism of electroweak symmetry breaking (EWSB). We will not discuss *how* this happens here, but for our purposes, suffice it to say that the Higgs field is a complex scalar $SU(2)_L$ -doublet that acquires a vacuum expectation value (VEV) along its neutral component:

$$H = \begin{pmatrix} H^+ \\ H^0 \end{pmatrix} \xrightarrow{\text{EWSB}} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \langle h \rangle + h(x) \end{pmatrix}, \quad (1.15)$$

where we omit the components H^+ and H^0 that are eaten by the $W^\pm$ and Z bosons, respectively, in the unitary gauge.² We also write $h(x)$ to emphasize that it is the physical Higgs field (it creates and destroys Higgs particles), while $\langle h \rangle = 246$ GeV is a constant VEV. We then go from a total of four real degrees of freedom in the doublet to one physical Higgs boson and three Goldstone bosons that are “eaten” by the $W^\pm$ and Z bosons.

The SM contains three generations of fermions that come in either left-chiral or right-chiral varieties. We discuss chirality in more details in the next section, but for now, you can imagine these two types of fermions as being completely different particles. Left-chiral fermions are $SU(2)_L$ doublets, while right-chiral fermions are $SU(2)_L$ singlets. For

²This is a choice, but it is a convenient one for our purposes. In other gauges, the Goldstone bosons H^+ and $\text{Im}\{H^0\}$ should be explicitly included in calculations. In the unitary gauge, they are “eaten” by the $W^\pm$ and Z bosons when these become massive. Recall that massless vector bosons like the photon have only 2 polarization modes, while massive ones have 3, the additional one being the longitudinal mode. This longitudinal degree of freedom for W and Z is supplied precisely by the Goldstone bosons.

example, for a single generation of fermions,

$$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \quad Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad e_R, u_R, d_R. \quad (1.16)$$

We list all of the fields in the SM in Table 1, including their representations/charges under the different gauge groups. Note that $SU(2)_L$, we only need to list the third component of weak isospin T_3 : left-chiral fermions, $T_3 = +1/2$ for the upper component of the $SU(2)_L$ doublet and $T_3 = -1/2$ for the lower component of the $SU(2)_L$ doublet. Right-chiral fermions are $SU(2)_L$ singlets, so they have $T_3 = 0$.

After EWSB, hypercharge Y and weak isospin T_3 are (spontaneously) broken, but a specific linear combination of them remains conserved. That is the electric charge Q . The Gell-Mann-Nishijima relation gives

$$Q = T_3 + \frac{Y_W}{2}. \quad (1.17)$$

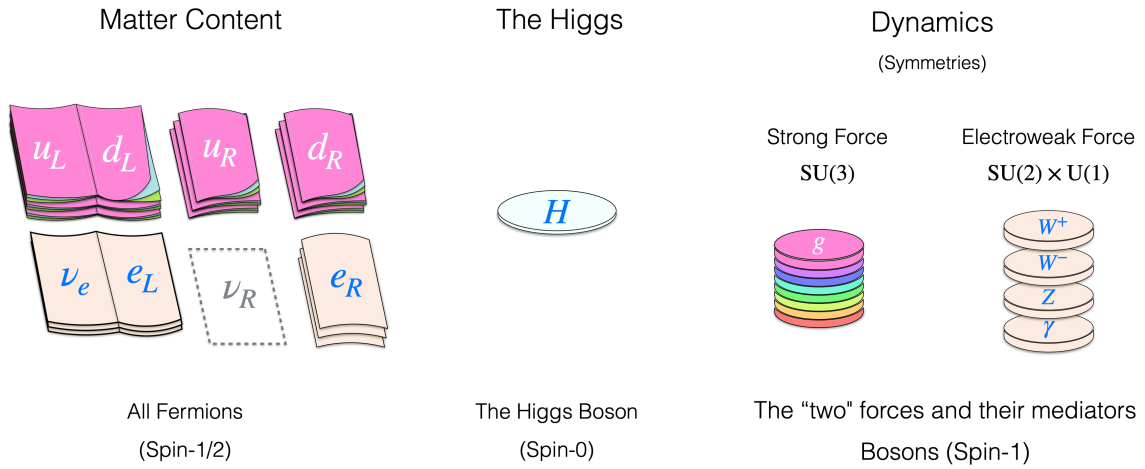
A few notes: 1) all SM fermions carry hypercharge, 2) neutrinos are the only fundamental fermions we know of that have no electric charge (the hypercharge and weak isospin cancel in Eq. (1.17)). While hypercharge and weak isospin are not conserved after EWSB, they are still useful to keep track of because the non-conservation is not random. It must be coming exclusively from the Higgs field, so for any given process that violates hypercharge or weak isospin conservation, we *must* have the Higgs field involved somehow, either by its VEV or by emitting or absorbing a physical Higgs boson (we expand on this in the next section).

Counting the number of matter fields gives: 3 generations $\times$ (2 left-chiral leptons + 2 left-chiral quarks + 1 right-chiral lepton + 2 right-chiral quarks) = 3 generations $\times$ 7 fields = 21 matter fields. Why not 24? Where did the three right-chiral neutrinos go? Well, we did not count them because we do not actually know if they exist or not. This is a major ambiguity in the SM: do ν_R states exist or not? How many of them are there? And how do they fit into the mechanism of neutrino masses? We know that all SM fermions combine into Dirac particles, but neutrinos are special and do not necessarily have to do so. These are all questions we explore in more detail in the lectures.

Particle	Color Charge	Electric Charge Q	Weak Isospin T_3	Hypercharge Y_W
Leptons				
ν_{eL}	singlet	0	$+\frac{1}{2}$	-1
e_L	singlet	-1	$-\frac{1}{2}$	-1
e_R	singlet	-1	0	-2
Quarks				
u_L^i	triplet ($i = R, G, B$)	$+\frac{2}{3}$	$+\frac{1}{2}$	$+\frac{1}{3}$
d_L^i	triplet ($i = R, G, B$)	$-\frac{1}{3}$	$-\frac{1}{2}$	$+\frac{1}{3}$
u_R^i	triplet ($i = R, G, B$)	$+\frac{2}{3}$	0	$+\frac{4}{3}$
d_R^i	triplet ($i = R, G, B$)	$-\frac{1}{3}$	0	$-\frac{2}{3}$
Gauge Bosons (after EWSB)				
g_{ij} (gluon)	octet	0	0	0
W^+	singlet	+1	+1	0
W^-	singlet	-1	-1	0
$Z = c_w W^0 - s_w B$	singlet	0	0	0
$\gamma = s_w W^0 + c_w B$	singlet	0	0	0
Higgs h	singlet	0	$+\frac{1}{2}$	+1

Table 1: Electric charge, color charge, weak isospin, and hypercharge assignments for all SM fields (for a single generation of fermions). For quarks, “triplet” refers to the three color assignments ($i = R, G, B$); for gluons, “octet” refers to the eight gluon states that are linear combinations of $i, j = R, G, B$.

Yet another way to visualize the field content of the SM is the following:



In this sketch, each page or element corresponds to a specific field or particle in the SM. The three generations are represented as separate layers or sections. The different color-charged quarks are indicated by pages of different colors. And finally, the left-handed and right-handed components versions of each fermion are represented as separate piles, with the right-handed neutrino pile left purposefully empty.



Symmetry breaking: For more details on the EWSB mechanism, see some of the reading material. What really matters for us is that the VEV breaks the electroweak symmetry down to electromagnetism:

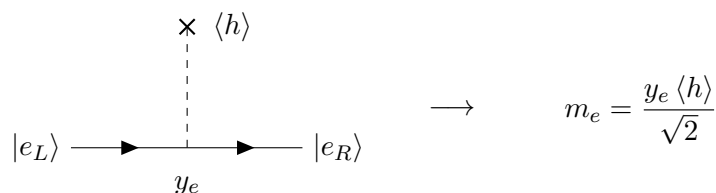
$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{EM}}, \quad (1.18)$$

and that this breaking has a fixed structure. Any electroweak-breaking effect in the SM *must* be proportional to the Higgs VEV v : the larger the VEV, the larger the breaking effects.

A very relevant example are fermion masses. As we will see, Dirac masses couple left-chiral and right-chiral fermions together, but these two types of fermions have different weak isospin and hypercharge (see Table 1; this is what we mean when we say the SM is *chiral*). Without the Higgs, fermions would be massless in the SM. However, the Higgs field provides exactly the right amount of symmetry breaking to generate this mass term. In the Lagrangian, this is described by the Yukawa interaction of electrons with the Higgs field. Using Eq. (1.15),

$$\mathcal{L}_{\text{Yukawa}} = -y_e \bar{L} H e_R + \text{h.c.} \longrightarrow -y_e \frac{\langle h \rangle}{\sqrt{2}} \bar{e}_L e_R + \dots = -m_e \bar{e}_L e_R + \dots \quad (1.19)$$

Diagrammatically, we can represent this as follows:



The cross at the end of the Higgs line emphasizes we needed to use the VEV rather than the quantum/particle of the Higgs field.

When an electron is created, it is the result of a superposition of left- and right-chiral states. The Dirac mass term maximally mixes e_L and e_R . If there were more interactions, between e_L and μ_R , for example, then the massive physical propagating degrees of freedom look like mixtures of e_L , e_R , and μ_R ; this is what we call *mass mixing*, in this case between fermions of different generations. It does not happen for charged leptons, but it does for quarks (giving rise to the CKM matrix) and neutrinos (giving rise to the PMNS matrix).

2 Chirality and Helicity

We just saw that SM fermions come in two types: left-chiral and right-chiral. Only the left-chiral particles participate in charged-current weak interactions. But what does *chirality* mean in the first place? How is it related to *helicity*? Which one do we measure in experiments? Which one is what we mean when we say “left-handed” or “right-handed” particles?

Chirality is as much a property of a particle as its charge or spin.³ In technical terms, chirality is determined by the representation of the fermion under the Lorentz group (the group of rotations, translations, and boosts: $SO(3,1) \simeq SU(2) \times SU(2)$). Left-chiral fermions transform as $(\frac{1}{2}, 0)$ and right-chiral fermions transform as $(0, \frac{1}{2})$. This may sound abstract, but it is no different from how we define spin: spin is determined by the representation of the particle under the rotation group.



Left-chiral particles have *negative* chirality, while right-chiral particles have *positive* chirality. We can define a chirality operator $\hat{\mathcal{C}}$ on *chiral states* as follows:

$$\hat{\mathcal{C}}|\psi_L\rangle = -|\psi_L\rangle, \quad \hat{\mathcal{C}}|\psi_R\rangle = +|\psi_R\rangle. \quad (2.1)$$

For four-component spinors, the chirality operator is just the γ^5 matrix:

$$\hat{\mathcal{C}} = \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \quad (2.2)$$

where $\mathbb{1}$ is the 2×2 identity matrix. The left-chiral and right-chiral components of a Dirac spinor ψ can be projected out using the projection operators

$$\psi_L = P_L\psi = \frac{1}{2}(1 - \gamma^5)\psi, \quad \psi_R = P_R\psi = \frac{1}{2}(1 + \gamma^5)\psi. \quad (2.3)$$

In the Weyl representation (or chiral basis), the left-chiral and right-chiral components of a Dirac spinor are simply the upper and lower two components, respectively:

$$\psi = \begin{pmatrix} \chi_L \\ \chi_R \end{pmatrix}, \quad (2.4)$$

with χ_L and χ_R being two-component Weyl spinors. In these notes, when we write ψ_L or ψ_R , we mean the left-chiral or right-chiral four-component spinor, namely, $P_L\psi$ or $P_R\psi$, respectively.

Chirality is important because it is what determines how a particle interacts with the weak force. In some sense, chirality is the “charge” of the weak interactions. In

³In reality, chirality may be seen as even more fundamental as it is defined by the mathematical representation of the fermion under the “external” *spacetime* symmetries that act on coordinates, namely the Poincare group of translation, rotation, and boost invariance. This is in contrast to the internal symmetries of the Standard Model that act on the internal degrees of freedom of the particles, such as color and flavor.

electromagnetism, chirality is irrelevant since these photons treat left-chiral and right-chiral particles the same way. Therefore, while in EM you can get away with working with the superposition

$$|e\rangle = \frac{|e_L\rangle + |e_R\rangle}{\sqrt{2}}, \quad (2.5)$$

in weak interactions, only the left-chiral component of the electron participates in charged-current weak interactions. The right-chiral component does not interact with the W boson, but it does interact with the Z boson in neutral-current weak interactions. This is because right-chiral fermions still carry hypercharge and because the Z boson contains an admixture of the hypercharge boson B .

Helicity Helicity is a related but distinct concept: it is the projection of the spin onto the direction of a particle's momentum. This is a *derived* and measurable quantity of particles, rather than an intrinsic property of a field. Helicity is defined as

$$\hat{h} = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}, \quad \hat{h} |\vec{p}, h\rangle = h |\vec{p}, h\rangle.$$

Inspecting this definition you should convince yourself that helicity is *not* a Lorentz-invariant quantity when the particle has a mass. We can always boost to a frame where the particle's momentum is reversed, which will flip the sign of the helicity. It also turns out that for massless particles, helicity and chirality coincide, which is why we often (sloppily) use the terms interchangeably.

Similarly to chirality, we call the helicity eigenstates

$$h = +\frac{1}{2} \text{ (right-handed)}, \quad h = -\frac{1}{2} \text{ (left-handed)}.$$

When trying to be more precise, we can refer to these as *left-helical* $|\nu_{-}\rangle$ and *right-helical* $|\nu_{+}\rangle$ states, as opposed to *left-chiral* $|\nu_L\rangle$ and *right-chiral* $|\nu_R\rangle$ states.



Helicity states are propagating states. Physical particles have well-defined helicity, unless they are at rest.

The relationship between helicity and chirality is fixed by the Dirac equation (you can prove this by writing the Dirac equation in the Weyl representation and solving for the spinor components). A useful, though slightly imprecise,⁴ way to relate the two bases is (we use ν from now on to refer to one of the neutrino flavors, but the same relations hold for any fermion):

$$|\nu_L\rangle = \frac{1}{\sqrt{2E}} \left(\sqrt{E+p} |\nu_{-}\rangle + \sqrt{E-p} |\nu_{+}\rangle \right), \quad (2.6)$$

$$|\nu_R\rangle = \frac{1}{\sqrt{2E}} \left(\sqrt{E-p} |\nu_{-}\rangle - \sqrt{E+p} |\nu_{+}\rangle \right), \quad (2.7)$$

⁴The states $|\nu_{L,R}\rangle$ are not really physical states, but a convenient proxy for the interaction eigenstates. We are omitting some details here about what exactly they mean, but this is the best I can do without using the full machinery of spinors.

where $E = \sqrt{p^2 + m^2}$ with $p = |\vec{p}|$. Another suggestive way to rewrite this is

$$|\nu_L\rangle = \frac{1}{\sqrt{2}} \left(\sqrt{1+\beta} |\nu_-\rangle + \sqrt{1-\beta} |\nu_+\rangle \right), \quad (2.8)$$

$$|\nu_R\rangle = \frac{1}{\sqrt{2}} \left(\sqrt{1-\beta} |\nu_-\rangle - \sqrt{1+\beta} |\nu_+\rangle \right), \quad (2.9)$$

where $\beta = \frac{p}{E}$ is the particle's velocity.

Let us consider the different regimes for these states. First, consider the stopped particle scenario, where $\beta \rightarrow 0$. In that case, the two helicities are maximally mixed together in the chiral states. This is not surprising: helicity states are fully scrambled for a particle at rest. Consequently, chirality is also fully scrambled. In the massless limit, $\beta \rightarrow 1$, the two helicities decouple completely and you will find that left-chiral states are equivalent to left-helical states, and that right-chiral states are equivalent to right-helical states. The relation for antiparticles is the same, with the replacement $\nu \rightarrow \bar{\nu}$ and $h \rightarrow -h$. The limits are then $|\bar{\nu}_L\rangle \rightarrow |\bar{\nu}_+\rangle$ and $|\bar{\nu}_R\rangle \rightarrow |\bar{\nu}_-\rangle$ as $\beta \rightarrow 1$ (i.e., positive-helicity antiparticles asymptote to left-chiral antiparticles in the ultra-relativistic limit).

Now, consider the ultra-relativistic expansion $\beta \rightarrow 1 - \frac{m^2}{2E^2}$:

$$|\nu_L\rangle \simeq |\nu_-\rangle + \frac{m}{2E} |\nu_+\rangle, \quad (2.10)$$

$$|\nu_R\rangle \simeq \frac{m}{2E} |\nu_-\rangle - |\nu_+\rangle, \quad (2.11)$$

Alternatively, we may choose to invert these relations,

$$|\nu_-\rangle \simeq |\nu_L\rangle + \frac{m}{2E} |\nu_R\rangle, \quad (2.12)$$

$$|\nu_+\rangle \simeq \frac{m}{2E} |\nu_L\rangle - |\nu_R\rangle. \quad (2.13)$$



Pion decay and helicity suppression. The diagram at the very beginning of these notes shows a pion decaying into a positron and a neutrino. This process is suppressed by the electron mass precisely because the correct helicity assignment required to conserve total angular momentum requires us to use the “wrong” helicity state of the left-chiral electron. We comment more on how to think about what states are produced in pion decay in Section 4

Discrete Symmetries. Three important discrete transformations in QFT can help us understand the roles of chirality and antiparticles: parity, time reversal, and charge conjugation. Their action on several relevant quantities is summarized below:

	Parity ($\hat{P}$)	Time reversal ($\hat{T}$)	Charge conjugation ($\hat{C}$)
Position	$\vec{x} \rightarrow -\vec{x}$	$\vec{x} \rightarrow \vec{x}$	$\vec{x} \rightarrow \vec{x}$
Time	$t \rightarrow t$	$t \rightarrow -t$	$t \rightarrow t$
Momentum	$\vec{p} \rightarrow -\vec{p}$	$\vec{p} \rightarrow -\vec{p}$	$\vec{p} \rightarrow \vec{p}$
Spin	$\vec{S} \rightarrow \vec{S}$	$\vec{S} \rightarrow -\vec{S}$	$\vec{S} \rightarrow \vec{S}$
Charge	$Q \rightarrow Q$	$Q \rightarrow Q$	$Q \rightarrow -Q$
Chirality	$ \psi_{L,R}\rangle \rightarrow \psi_{R,L}\rangle$	$ \psi_{L,R}\rangle \rightarrow \psi_{L,R}\rangle$	$ \psi_{L,R}\rangle \rightarrow \bar{\psi}_{L,R}\rangle$
Helicity	$h \rightarrow -h$	$h \rightarrow h$	$h \rightarrow h$

Note that the action of $\hat{C}$ on states is to flip their charges (particle to antiparticle). The action of $\hat{P}$ is to reverse the spatial coordinates, effectively flipping the chirality of the field.⁵ Finally, $\hat{T}$ flips the sign of both the momentum and spin, but leaves both helicity and chirality unchanged.

The combination $\widehat{CPT}$ is important. It turns out that Lorentz invariance and CPT invariance go hand in hand. Acting each transformation in order on our chiral states gives

$$\widehat{CPT} |\psi_L, \vec{p}\rangle = \widehat{CP} |\psi_L, -\vec{p}\rangle = \hat{C} |\psi_R, \vec{p}\rangle = |\bar{\psi}_R, \vec{p}\rangle, \quad (2.14)$$

$$\widehat{CPT} |\psi_R, \vec{p}\rangle = \widehat{CP} |\psi_R, -\vec{p}\rangle = \hat{C} |\psi_L, \vec{p}\rangle = |\bar{\psi}_L, \vec{p}\rangle. \quad (2.15)$$

What this means is that CPT transformations take us from left-chiral particles to right-chiral antiparticles. Therefore, because of CPT invariance, these two particles must be two sides of the same coin. They come from the same field: a chiral field excites particles as well as its antiparticles with opposite chirality. For instance, the chiral field $e_L(x)$ creates $|e_L\rangle$ and $|\bar{e}_R\rangle$ states. You cannot have one without the other.



This is why the original SM could get away without including right-handed neutrino fields: ν_L^e already creates both $|\nu_L^e\rangle$ and $|\bar{\nu}_R^e\rangle$ states. The former produces *electrons* in charged current interactions, while the latter produces *positrons* in charged current interactions.

⁵One can show this without resorting to the concept of spin and helicity by noting that the spatial component Dirac matrices γ^i change sign under parity, while the time component γ^0 does not; in that case, $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ changes sign, and therefore we can exchange $P_L \leftrightarrow P_R$ under parity transformations.



CPT as a “complex” Lorentz transformation. A Lorentz boost in the x direction is described by the matrix

$$\Lambda(\eta) = \begin{pmatrix} \cosh(\eta) & -\sinh(\eta) & 0 & 0 \\ -\sinh(\eta) & \cosh(\eta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (2.16)$$

where η is the rapidity parameter related to the boost velocity by $\beta = \tanh(\eta)$. Usually we think of β and η as real parameters, but show that applying a boost with $\eta = i\pi$ and a subsequent rotation of π around the x axis, this matrix corresponds to a $\widehat{PT}$ transformation as far as the coordinates are concerned. Knowing that $\widehat{C}$ acts to transform a particle to an antiparticle and that, a particle traveling backwards in time is equivalent to an antiparticle traveling forwards in time (this is the Feynman-Stueckelberg interpretation of antiparticles), conclude that the combination $\widehat{CPT}$ is analogous to this “complexified” Lorentz-invariance.

3 Fermion Masses and Neutrinos

Majorana vs Dirac fermions. We saw that Dirac fermions are superpositions of two chiral states, $|\psi_L\rangle$ and $|\psi_R\rangle$. What this implies is that they require the existence of two chiral fields: one excites $|\psi_L\rangle$ and $|\bar{\psi}_R\rangle$ while the other excites $|\psi_R\rangle$ and $|\bar{\psi}_L\rangle$. Diagrammatically, you can represent a Dirac mass as follows:

$$|\psi_L\rangle \longrightarrow \text{---}\times\text{---} \longrightarrow |\psi_R\rangle$$

m_ψ

As before, if the mass comes from the Higgs VEV (as in the SM), the mass insertion (the cross) is replaced by an interaction with the Higgs background, guaranteeing gauge invariance. In Eq. (2.10), we saw how masses scramble chiralities in the propagating states of definite helicity, requiring at least two chiral fermions. However, Dirac masses are not the only way to combine two chiralities together.

If we allow for a fermion to couple to its own antiparticle, we can mix two chiral states *from the same field* with a Majorana mass. That is, Majorana masses couple the fermion $|\psi_L\rangle$ and its own antiparticle partner, $|\bar{\psi}_R\rangle$. Diagrammatically, you can represent a Majorana mass as follows:

$$|\psi_L\rangle \longrightarrow \text{---}\times\text{---} \longleftarrow |\bar{\psi}_R\rangle$$

μ_ψ

Note the clashing arrows. They indicate that the charges associated with the fermion ψ are not conserved by a Majorana mass. In the example above, if the fermion carried electric charge Q , the Majorana mass would violate it by two full units, $\Delta Q = 2Q$. This explains why, with the exception of neutrinos, all SM fermions must be Dirac particles.

As far as we know, neutrinos are the only elementary particles that can acquire a Majorana mass.⁶ For now, let us ignore the potential existence of ν_R . A SM neutrino state $|\nu_L\rangle$ combining with its own antiparticle $|\bar{\nu}_R\rangle$ would give us the two helicity states of a Majorana neutrino:

$$|\nu_-\rangle \simeq |\nu_L\rangle + \frac{m_\nu}{2p} |\bar{\nu}_R\rangle, \quad |\nu_+\rangle \simeq -|\bar{\nu}_R\rangle + \frac{m_\nu}{2p} |\nu_L\rangle. \quad (3.1)$$

Note it mixes two states with opposite quantum numbers. This brings up an important question: where should we put the bar ($\bar{}$)? Should it go on $\bar{\nu}_-$, $\bar{\nu}_+$, or both? This ambiguity is a reflection of the fact that the helicity states mix particle and antiparticle states. Strictly speaking, they are neither purely particles nor antiparticles.

i

This is why you will often hear that

Majorana particles are their own antiparticles.

This sentence is unfortunately quite confusing because it omits the fact that there are still two different particles produced from the Majorana field, namely the two different helicity eigenstates. From the relation in Eq. (3.1), you can see that ultra-relativistic ν_- states will interact mostly as a *neutrino*, while ν_+ will interact mostly as an *antineutrino*. This is why we can still talk about neutrinos and antineutrinos in the context of Majorana masses: by convention, we call the *left-helical* states by neutrinos and the *right-helical* by antineutrinos. In the non-relativistic case, this distinction is far less useful, and both helicity states interact equally as neutrinos and antineutrinos.

i

How many different degrees of freedom does a Dirac fermion have? Well, firstly, we already know that $|e_L\rangle$ and $|e_R\rangle$ will produce two independent states each, so $n_{\text{dof}} = 4$. The Dirac field gives

$$\psi(x)_{\text{Dirac}} = \psi_L(x) + \psi_R(x) \text{ creates } |\psi_L\rangle, |\bar{\psi}_R\rangle, |\psi_R\rangle, \text{ and } |\bar{\psi}_L\rangle. \quad (3.2)$$

In weak interactions, however, the Dirac field only shows up in the combination $\hat{P}_L \psi(x)_{\text{Dirac}} = \psi_L(x)$, so it can only produce the states $|\psi_L\rangle$ and $|\bar{\psi}_R\rangle$.

This should be contrasted with a Majorana or Weyl fermion. There, only one chirality is present, leading to half as many degrees of freedom, $n_{\text{dof}} = 2$:

$$\psi_L(x) \text{ creates } |\psi_L\rangle \text{ and } |\bar{\psi}_R\rangle, \quad (3.3)$$

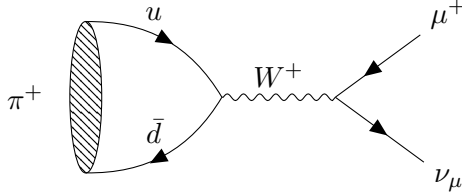
and similarly for ψ_R . But crucially, ψ_R and ψ_L are independent fields and need not both exist. A Weyl fermion has no Majorana mass, but has the same number of degrees of freedom as a Majorana fermion.

⁶If dark matter is a fermionic particle, it may also be a Majorana particle since dark matter should be electrically neutral to have escaped detection so far.

4 Dirac-Majorana Neutrino Confusion Theorem

Given how different the Dirac and Majorana cases are, shouldn't we be able to distinguish them experimentally? After all, Dirac neutrinos have twice as many degrees of freedom as Majorana neutrinos. The problem, however, is that the propagating states of definite helicity are almost always ultra-relativistic, and therefore, contain very little amount of the “wrong” chiral/interaction state (compare Eq. (2.12) with Eq. (3.1), they are identical up to corrections of order m/E). In the Dirac case, the wrong-chiral content of ν_- is $|\nu_R\rangle$, which does not interact with the weak force at all, while in the Majorana case, the wrong-chiral content of ν_- is $|\bar{\nu}_R\rangle$, which does interact with the weak force, producing antileptons in charged-current interactions as opposed to leptons. This similarity is known as the *Dirac-Majorana confusion theorem*: the two cases are experimentally indistinguishable in the ultra-relativistic limit.

We can ask how hard it would be to distinguish these two scenarios by observing the wrong-chirality content of the neutrino. We start with a neutrino source, say, pion decay $\pi^+ \rightarrow \mu^+ + \nu_\mu$:



Because this is a weak interaction process, this decay proceeds through the interactions of a right-handed anti-muon $|\bar{\mu}_R\rangle$ and a left-handed muon neutrino $|\nu_L\rangle$.⁷ Now, we know that the propagating degrees of freedom are helicity states, and not chirality states. So, when the pion decays, we must project the interaction states onto the propagating degrees of freedom to find the actual helicity states produced. Schematically, the branching ratios for pions to decay into the two states are determined by the overlap of the interaction states with the physical helicity states:⁸

$$\mathcal{B}(\pi^+ \rightarrow \mu^+ \nu_-) \propto |\langle \nu_- | \nu_L \rangle|^2 \simeq 1 - \frac{m^2}{4E^2}, \quad (4.1)$$

$$\mathcal{B}(\pi^+ \rightarrow \mu^+ \nu_+) \propto |\langle \nu_+ | \nu_L \rangle|^2 = \frac{m^2}{4E^2}. \quad (4.2)$$

The pion decays mostly to $|\nu_- \rangle$ and a tiny amount of $|\nu_+ \rangle$. To emphasize: the pion does *not* decay to the superposition $|\nu_L \rangle$, but rather, each time the pion decays, it produces either $|\nu_- \rangle$ or $|\nu_+ \rangle$ with the aforementioned proportions. A similar thing happens when we study the difference between flavor and mass eigenstates in neutrino oscillations.

If neutrinos are Majorana, this small amount of $|\nu_+ \rangle$ is quite significant. Consider the left-helical states $|\nu_- \rangle$. These are almost identical to left-chiral neutrinos: $|\nu_L \rangle$, which

⁷For now, we will drop the flavor subscript ν_μ and simply write ν for simplicity.

⁸Note that there is no interference between the two amplitudes because the two helicity states are orthogonal.

through the weak force, scatter on nucleons as follows:

$$\nu_L + n \rightarrow \mu^- + p^+, \quad \sigma(\nu_- + n \rightarrow \mu^- + p^+) \propto |\langle \nu_- | \nu_L \rangle|^2 \simeq 1 - \frac{m^2}{4E^2}. \quad (4.3)$$

Now, the right-helical state $|\nu_+\rangle$ for Majorana neutrinos is almost identical to $|\bar{\nu}_R\rangle$, which also interacts with the weak force, but it does so as an antiparticle. The scattering on nucleons is then strikingly different:

$$\bar{\nu}_R + p^+ \rightarrow \mu^+ + n, \quad \sigma(\bar{\nu}_+ + p^+ \rightarrow \mu^+ + n) \propto |\langle \nu_+ | \bar{\nu}_R \rangle|^2 \simeq 1 - \frac{m^2}{4E^2}. \quad (4.4)$$

This is why we refer to ν_+ as an antineutrino in the Majorana case.

Now, using only the helicity eigenstates, if you keep track of lepton number in these reactions, you will find that there are four possible chains of events, shown in Fig. 2. The first conserves L overall,

$$\pi^+ \rightarrow \mu^+ + \nu_- \quad \text{followed by} \quad \nu_- + n \rightarrow \mu^- + p^+ \quad (\Delta L = 0). \quad (4.5)$$

Overall, this process starts with a pion, produces a μ^+ and a ν_- . The latter has no well-defined lepton number (it is a combination of a neutrino and an antineutrino), so to determine the overall lepton number in the full process, we will ask what charged lepton gets produced once it scatters, which, in this case, is a μ^- . So, starting with the pion, which had no lepton number, we ended up with a μ^+ and a μ^- , which have opposite lepton numbers and therefore, no net gain or loss of L took place. Note that the neutron is there just to serve as a target for the neutrino and also does not change the overall lepton number.

Now, let us consider a chain of events that violates L ,

$$\pi^+ \rightarrow \mu^+ + \nu_+ \quad \text{followed by} \quad \nu_+ + p^+ \rightarrow \mu^+ + n \quad (\Delta L = 2). \quad (4.6)$$

Overall, this second chain sees the production of two μ^+ particles: one from pion decay, and the other from the neutrino scattering on a nucleon. This chain of events is much less likely because we must pay the price of $\langle \nu_L | \nu_+ \rangle \simeq \mathcal{O}(m^2/E^2)$ in pion decay.

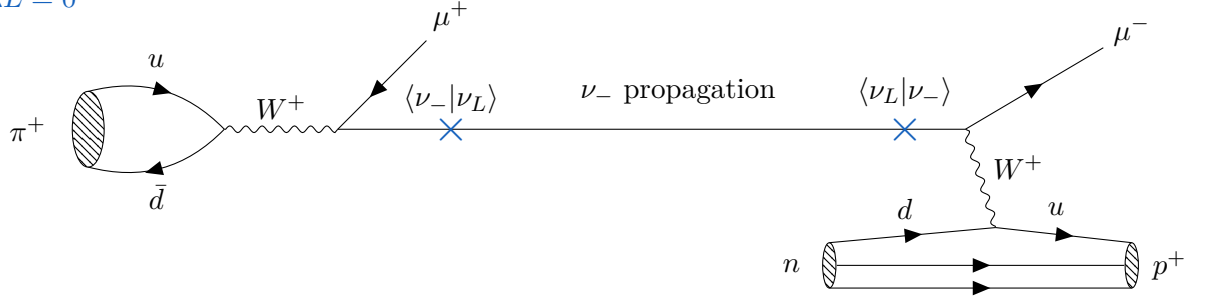
The two remaining possibilities come from the different ways in which the helicity states can scatter on nucleons. For the L violating one, you could have also asked that ν_- interact as a right-chiral *antineutrino* $|\bar{\nu}_R\rangle$. This is again suppressed by an overall factor of $m^2/4E^2$. And the last possibility, and least important one, is to require that the pion decay to ν_+ and but force it to interact as a left-chiral *neutrino* $|\nu_L\rangle$. In this case, the overall suppression is of order $m^4/16E^4$, so completely negligible.

So, how likely are we to observe such lepton number violating processes and prove neutrinos are Majorana particles? For a typical accelerator experiment, we get

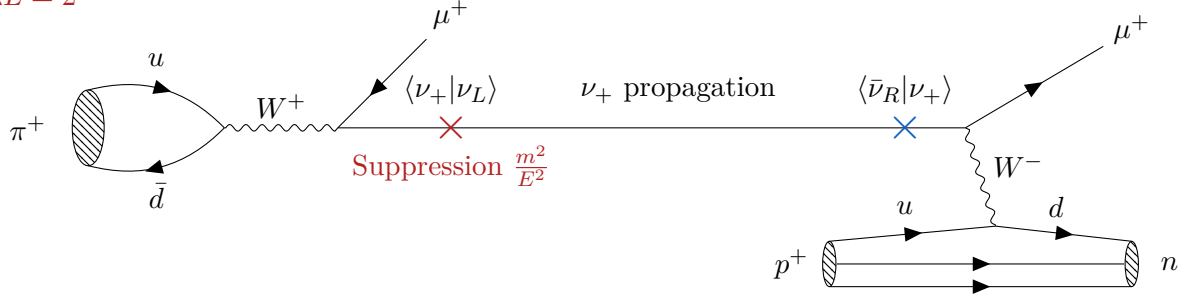
$$\frac{m^2}{4E^2} \simeq 2.5 \times 10^{-21} \left(\frac{m_\nu}{0.1 \text{ eV}} \right)^2 \left(\frac{1 \text{ GeV}}{E_\nu} \right)^2. \quad (4.7)$$

This is telling us that you would need to detect about 10^{21} neutrinos in total in order to see a single μ^+ in the detector in beam made exclusively from π^+ decay, which is completely unrealistic.

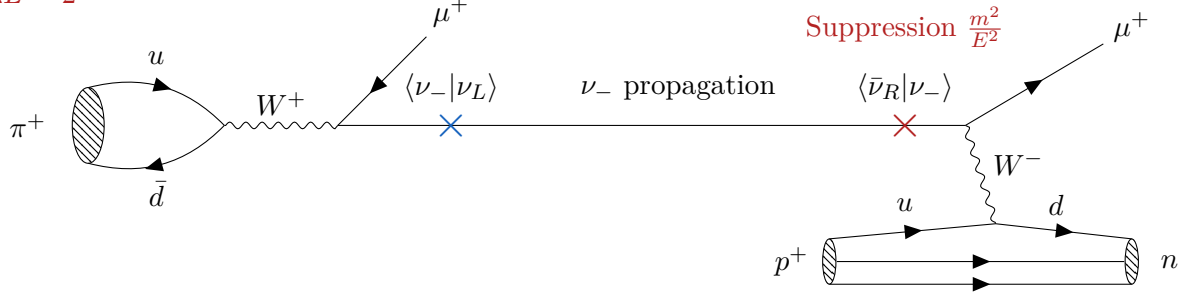
$\Delta L = 0$



$\Delta L = 2$



$\Delta L = 2$



$\Delta L = 0$

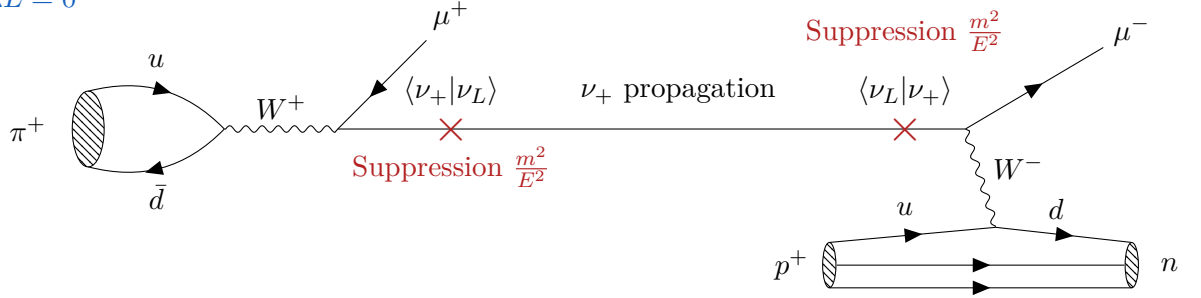


Figure 2: Pion decay to neutrinos followed by neutrino scattering on a nucleon target for different possibilities under a Majorana neutrino scenario.